Epsilon FET

User manual & mixing guide

A circuit modeled FET compressor for shaping vocals, bass, drums, and buses.



ACTUAL v0.2.0 PANEL / DRUMS - PARALLEL SMASH

VST3 / AUDIO UNIT / STANDALONE

Intel and Apple Silicon Mac builds. The guide covers every knob and switch, all nine factory presets, and settings you can try in a mix.

START WITH INPUT

Drive the compressor with Input. Match the listening level with Output.

Find what you need

Start with installation and one listening pass. Tap a row to jump to that page.

PAGE	TOPIC	USE IT FOR
03	**Install on macOS**	Finder copy, FL Studio scan, AU and standalone
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05	**Your first listening pass**	Set gain reduction and compare at equal loudness
06	**Input and Output**	Drive, makeup level, and output color
07	**Attack and Release**	What the 1 to 7 scale does and how fast it reacts
08	**Ratio and Mix**	Five ratio buttons and the dry/wet blend
09	**VU meter and modes**	GR, +4, +8, and OFF
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17	**Measured behavior**	Intel validation, latency, and model reference

TWO THINGS TO REMEMBER

Higher Attack and Release numbers mean faster timing. In GR mode, a needle at -5 means about 5 dB of gain reduction in the compressed path.

Install on macOS

Use the compiled v0.2.0 ZIP on your Intel or Apple Silicon Mac. The build targets macOS 12 or later.

1. **Unzip the download**

   Double-click **EpsilonFET-v0.2.0-macOS.zip** in Finder. If you downloaded the outer GitHub Actions artifact, unzip it first and open the macOS ZIP inside.

2. **Copy the whole VST3 bundle**

   In Finder, press **Shift + Command + G** and enter `~/Library/Audio/Plug-Ins/VST3/`. Create the folder if needed. Copy **Epsilon FET.vst3** into it as a single item.

3. **Scan in FL Studio**

   Open **Options > Manage plugins**, then choose **Find installed plugins**. Search the results for **Epsilon FET**. If the first scan misses it, enable **Verify plugins** and scan again.

4. **Insert the effect on a Mixer track**

   Route your vocal, bass, or drum track to a Mixer insert. Open an empty effect slot, choose **More plugins** if needed, then select the **VST3** entry for Epsilon FET.

PACKAGE ITEM	LOCATION OR ACTION	WHEN TO USE IT
Epsilon FET.vst3	~/Library/Audio/Plug-Ins/VST3/	FL Studio and other VST3 hosts.
Epsilon FET.component	~/Library/Audio/Plug-Ins/Components/	Audio Unit hosts such as Logic.
Epsilon FET.app	Open the app in Finder. Use Options > Audio/MIDI Settings to choose devices.	Standalone processing through your audio interface. Use headphones if you are monitoring a mic.

ONE FORMAT PER PROJECT

For a new FL Studio project, use the VST3 entry. When reopening a saved session, keep the same plugin format it was saved with.

Mac build and download | Image-Line plugin installation guide

Front panel map

Tap a numbered marker to open its instructions. This panel shows Drums - Parallel Smash loaded.



CONTROL	PAGE	CONTROL	PAGE
01 Preset menu	11	**08** Release	07
02 Power	10	**09** Mix	08
03 VU needle	09	**10** Stereo Link	10
04 GR / +4 / +8 / OFF	09	**11** Studio 4x / Eco 2x	10
05 Input	06	**12** Ratio 4 / 8 / 12 / 20 / ALL	08
06 Output	06	**13** X-RAY	15
07 Attack	07		

MOUSE GESTURES

Drag a knob up to increase its value and down to decrease it. Read the value under the knob. Double-click to restore its factory default. Hover over a control for its tooltip.

Your first listening pass

Loop a loud phrase on a vocal or drum track. Work at a comfortable monitoring level and watch the Mixer peak meter.

1. **Start from Init**

   Select **Init** in the preset menu. Confirm **POWER** is lit, the meter is on **GR**, and **Mix** is 100%. Use Ratio 4 and Attack 3. Start Release at 4 or 5 for a smoother example.

2. **Set how much it compresses**

   Turn **Input** up until the GR needle moves left on the louder words or hits. Aim for about **3 to 5 dB** on peaks in this example. If the track is already hot, turn Input down to get there.

3. **Shape the front of the sound**

   Try Attack 2 and then 5 on the same phrase. The lower number keeps more of the initial consonant or drum crack. The higher number catches it sooner. Recheck GR after changing timing.

4. **Set the recovery between hits**

   Watch the needle approach 0 in gaps. Move Release up a number if gain stays down into the next hit. Move it down if the tail surges or the bass gets grainy.

5. **Match loudness and compare**

   Toggle **POWER** on and off over the same phrase. Adjust **Output** with Power on until the two versions sound equally loud. Check peaks on the FL Studio Mixer meter and keep headroom.

6. **Blend for parallel compression**

   If the full effect flattens too much, try **Mix 60 to 80%**. This lets more dry transients through. Match loudness again before deciding whether to keep the change.

> **WORKED EXAMPLE ON A VOCAL**
>
> Say the loudest word shows 1 dB of GR at Input 0. Raise Input until the same word shows about 4 dB. Suppose it takes +6 dB on this recording. Now toggle Power over that word and its tail, and trim Output by ear. The two versions should feel equally loud before you judge how the tail sits.

Input and Output

Input sets drive into the compressor. Output sets the level feeding its final amplifier and transformer.



05 Input

-24 TO +24 dB / DEFAULT 0 dB

Input changes the level going from the input transformer into the FET and its feedback detector. Each ratio network has its own fixed threshold behavior. Raising Input generally makes the needle show more GR.

Try it. Loop your loudest phrase. Turn Input up or down until it hits the GR you want. If a loud track already compresses at 0 dB, reduce Input.



06 Output

-24 TO +24 dB / DEFAULT 0 dB

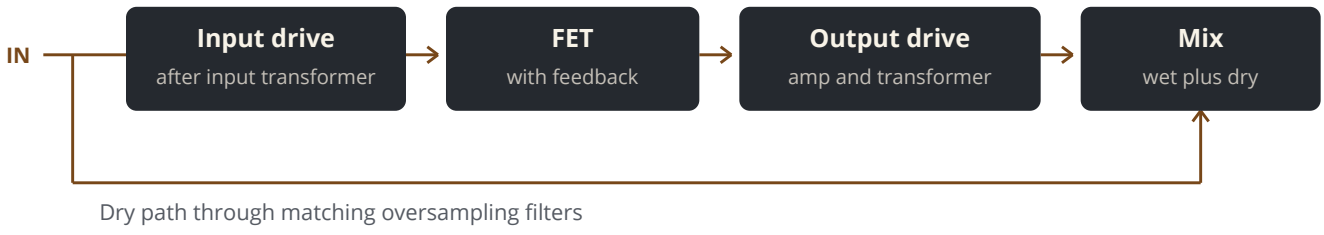
Output applies gain after the detector tap and before the modeled class-A output stage and output transformer. Use it to match the bypassed loudness. Higher drive here can add color or raise peaks.

Try it. After setting Input, toggle Power and compare the same phrase. Adjust Output while Power is on. Use the Mixer fader for a final trim after the dry/wet blend.

EXAMPLE ON A LOUD VOCAL

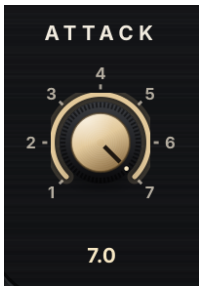
If GR is around 8 dB and you want 4 dB, reduce **Input** until the GR reading is closer. Then adjust **Output** to level match. If the input transformer itself is overdriven, reduce the source level or a gain stage before Epsilon FET.

Wet and dry paths



Attack and Release

Both knobs run from 1 (slowest) to 7 (fastest). The numbered scales are continuous, and the display shows one decimal place.

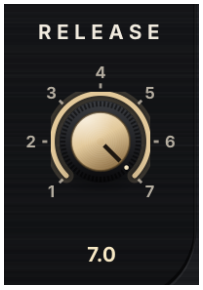


07 Attack

1 TO 7 / DEFAULT 3

Sets how quickly the detector charges and the FET turns a rising signal down. Lower numbers leave more of the initial hit. Higher numbers help catch sharp peaks.

Try it. Use 1 to 3 for snare punch or 5 to 7 to soften sharp consonants. The fastest setting has a nominal detector time constant near 20 µs.



08 Release

1 TO 7 / DEFAULT 7

Sets how quickly gain comes back after the detector sees less signal. A fast release adds movement. A slow release holds the level down longer.

Try it. Start near 4 for a smooth vocal. Speed it up if it holds down the next word. Slow it down if bass or room tails surge or buzz.

Timing measured on the circuit model

SETTING	1	2	3	4	5	6	7
Attack	760 µs	289 µs	198 µs	147 µs	125 µs	108 µs	102 µs
Release	1088 ms	680 ms	409 ms	244 ms	146 ms	87 ms	52 ms

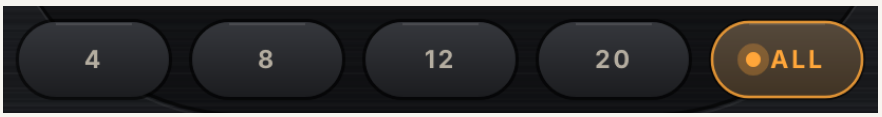
Original circuit-engine test. Ratio 4, 1 kHz level step from -40 to -12 dBFS and back, measured at 63% of the gain change or recovery. Timing in a mix depends on the source and ratio. The detector time constant and the measured audio response describe different things.

RELEASE AND THE BEAT

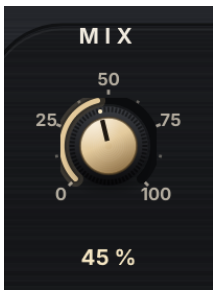
At 130 BPM, sixteenth notes are about 115 ms apart. If Release 4 holds down the next hit, try 6 and watch the needle recover. On sustained bass, listen for extra grit as you speed it up.

Ratio and Mix

Choose the compression behavior with the five Ratio buttons. Set the amount you hear with the Mix knob.



BUTTON	BEHAVIOR	START HERE
4	Nominal 4:1. The lowest threshold among the four numbered ratios. A good first setting for moderate control.	Lead vocal, bass, or gentle bus work.
8	Nominal 8:1. Stronger control above a higher threshold.	Forward vocals or snare that needs firmer peak control.
12	Nominal 12:1. Firmer peak control with a higher threshold again.	Sharp syllables, percussion, or a second stage of compression.
20	Nominal 20:1. The steepest numbered ratio and highest numbered threshold.	Catch harder peaks. Use a dedicated limiter when a true-peak ceiling is required.
ALL	All-buttons mode. The modeled ratio network lowers the threshold and speeds the attack, with a different bias and a more aggressive response.	Room mics and parallel drums. Start with lower Input, then bring Mix up to taste.



09 Mix

0 TO 100% WET / DEFAULT 100%

Blends dry and compressed audio after Output. Use 0% for dry, 100% for wet, or 45% for 45% wet and 55% dry. Both paths use matching oversampling filters for timing alignment.

Try it. Dial in a strong drum sound at 100%, then drop Mix to about 45% for parallel compression. Recheck loudness and peak headroom.

RECHECK INPUT WHEN YOU SWITCH RATIOS

Because the numbered ratios also move the effective threshold, a higher ratio can show less GR at the same Input level. Use the GR meter and turn Input to the amount you want.

VU meter and modes

A new instance starts in GR. The VU needle follows the signal more slowly than a digital peak meter.



Choose a mode below the needle.

MODE	WHAT THE NEEDLE SHOWS
GR	Gain reduction in the wet circuit. It rests near 0 with no compression. A reading of -5 means about 5 dB of attenuation.
+4	Output level after the blend. 0 VU is the modeled +4 dBu reference, approximately -18 dBFS peak for a steady sine (-21 dBFS RMS).
+8	Output level with 0 VU referenced 4 dB higher, around -14 dBFS peak for a steady sine. It reads 4 dB lower than +4 for the same output.
OFF	Parks the needle. The selected meter mode is remembered with the session. All four modes leave the sound unchanged.

Three useful readings

1. **Set Input with GR**
   Watch how far the needle falls on loud words and how close it gets to 0 between them. When Link is off, the GR readout follows the more compressed detector.

2. **Check loudness with +4**
   Use the same passage while toggling Power. The +4 reading and your ears can help match output. Use the FL Studio Mixer peak meter for digital headroom.

3. **Check Power and Mix separately**
   The detector keeps running at Mix 0% or with Power off. GR can still move while the output is dry. The Power lamp and Mix readout tell you what you are hearing.

EXAMPLE

If GR swings to -6 and returns near 0 between hits, you are seeing roughly 6 dB of wet-path reduction on those hits. At a low Mix setting the change in the combined output will be smaller.

Power, Link, and Quality

Use Power for comparisons, Link for stereo behavior, and Quality to set the internal processing rate.

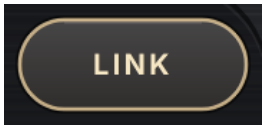


02 Power

ON / OFF / DEFAULT ON (LIT)

Lit means the processed blend is active. Off fades to the aligned dry path at the current quality setting, keeping the reported latency. The circuit and detector still run.

Try it. Match loudness with Output, then click Power several times over the same loop. For a CPU check, disable the entire effect slot in the host.

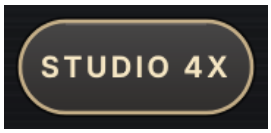


10 Link

ON / OFF / DEFAULT ON

Click LINK to toggle. When outlined, it shares one detector voltage between the stereo channels, weighted toward the stronger detector. With Link off, left and right compress independently.

Try it. Leave it on for a stereo drum or mix bus to steady the image. Try off on a room pair for movement and listen for the image shifting.



11 Quality

STUDIO 4X / ECO 2X / DEFAULT STUDIO 4X

Studio runs the model at four times the session sample rate. Eco runs it at twice the rate and reduces CPU use. Click the button to switch.

Try it. Use Eco on busy Intel Mac sessions and compare Studio on a short bounce. Change quality with playback stopped because the circuit and latency are reset.

RESIZE AND READ THE FOOTER

Drag the bottom-right handle to resize at the fixed panel aspect ratio. The standard view footer shows the plugin version and reported latency. Close the window or turn off X-RAY to reduce graphics work.

Presets and automation

A new instance starts at Init. Click the preset name at the top of the panel to load one of nine factory starting points.



PRESET	INPUT	OUTPUT	ATTACK	RELEASE	RATIO	MIX
Init	0 dB	0 dB	3	7	4	100%
Vocal - Classic	+8 dB	-3 dB	3	7	4	100%
Vocal - Upfront	+12 dB	-6 dB	5	6	8	100%
Bass - Even	+6 dB	-2 dB	2	4	4	100%
Drums - Parallel Smash	+14 dB	-8 dB	7	7	ALL	45%
Room Mics - All In	+14 dB	-9 dB	7	7	ALL	100%
Snare - Snap	+6 dB	-2 dB	1	7	8	100%
Mix Bus - Gentle	+3 dB	-1 dB	1	3	4	60%
Peak Limiter	+6 dB	-2 dB	7	5	20	100%

What loading a preset changes

A preset sets the six values in the table. Power, Link, Quality, meter mode, and X-RAY keep their current settings. Init resets the same six values. The menu can still show the last factory name after you edit the knobs.

Recall and automation in FL Studio

Save the FL Studio project to recall the plugin state. For a reusable preset, use **Save preset as** in the wrapper menu. To automate a control, move it once and use **Tools > Last tweaked > Create automation clip**, or select it from the wrapper parameter list. Try a small Mix change between verse and chorus, then level match both sections.

PARAMETERS THE HOST CAN AUTOMATE

Input, Output, Attack, Release, Ratio, Mix, Stereo Link, Quality, and Bypass (the Power switch). Meter mode and X-RAY are saved display settings.

Vocals

Loop the loudest phrase and a quieter one. The GR targets below refer to what you see on loud words, after adjusting Input for the recording.

Smooth lead vocal

START FROM VOCAL - CLASSIC

```
RATIO 4 ATTACK 3 RELEASE 4-5 MIX 100% GR 2-5 dB
```

Load the preset, then slow Release from 7 to about 4 or 5. Set Input for light movement on normal words and about 2 to 5 dB on louder ones. Match Output to the bypassed phrase.

Listen for the quieter ends of words staying audible. Try Attack 2 if consonants lose definition.

Forward rap or ad-lib

START FROM VOCAL - UPFRONT

```
RATIO 8 ATTACK 5 RELEASE 6 MIX 75-100% GR 4-7 dB
```

Set Input until the loud syllables stay in place. Blend down toward 75% if the full effect feels too dense. Match the level again and compare in the beat.

Listen for breaths and room sound rising between bars. Move Release toward 5 or back down Input if they jump forward.

Stereo vocal stacks

START FROM INIT

```
RATIO 4 ATTACK 2 RELEASE 4 MIX 70-85% GR 1-3 dB
```

Put Epsilon FET on the vocal bus and leave Link on. Set Input so the loudest stack leans into the compressor without ducking the entire group.

Listen for a steady center and clear ad-libs on the sides. If sharp S sounds trigger too much compression, address them with a de-esser earlier in the chain.

808s and bass

Test low and high notes in the same pattern. Match levels when comparing how the low end feels in the beat.

Even, rubbery 808

START FROM BASS - EVEN

```
RATIO 4 ATTACK 2 RELEASE 3-4 MIX 65-85% GR 1-3 dB
```

Start with the preset and lower Mix to bring back some dry shape. Trim Input so only the notes that jump out pull the needle left. Set Output so bypass and processed versions feel equally loud.

Listen for a rounded onset and stable sustain. If the next note stays held down, speed Release up one number and recheck the bass texture.

A little grit in parallel

START FROM INIT

```
RATIO 4 ATTACK 5-7 RELEASE 6-7 MIX 20-40% GR 3-5 dB
```

Raise Input with Mix temporarily at 100% so you can hear what the wet signal is doing. Then blend it down. Fast timing can make the FET follow part of the low-frequency waveform and add harmonics.

Listen for a controlled edge on small speakers. If the fundamental loses its roundness, slow Release toward 4 and reduce Mix.

Placement around distortion

PLACEMENT	WHEN IT HELPS
Before distortion	Even out big note-to-note level swings so they drive the next stage more consistently.
After distortion	Level the full distorted signal and compare how the 808 holds its place in the mix.

WHEN 300 TO 400 HZ IS THE PROBLEM

Epsilon FET turns the whole signal down when its detector reacts. To reduce only the extra punch near 300 to 400 Hz, pair it with an EQ or a dynamic EQ set to that band.

Drums and buses

Pick the envelope you want to keep. Use the wet path for weight, then add back the dry attack with Mix.

Snare with a clear crack

START FROM SNARE - SNAP

```
RATIO 8 ATTACK 1 RELEASE 7 MIX 100% GR 2-5 dB
```

Set Input on the loud hits. Use Attack 1 to keep as much of the hit as this fast compressor allows. Move Attack toward 4 if the snare still sticks out.

Listen for the hit and tail separately. Slow Release if the ring or bleed gets busy.

Parallel drums or room crush

START FROM DRUMS - PARALLEL SMASH

```
RATIO ALL ATTACK 7 RELEASE 7 MIX 30-45% GR 6-12 dB
```

Keep Link on for a stereo bus. Push Input until the wet channel adds weight and motion, then blend it under the dry kit. For a separate room track, Room Mics - All In starts at 100% wet.

Listen for cymbal wash and kick weight. Back down Input if every kick pulls the kit down. Slow Release a little if the cymbals breathe too fast.

Gentle mix bus movement

START FROM MIX BUS - GENTLE

```
RATIO 4 ATTACK 1 RELEASE 3 MIX 60% GR 1-2 dB
```

Leave Link on. Raise Input until the loudest section moves the needle slightly. Level match against Power off and compare the kick, vocal, and stereo center.

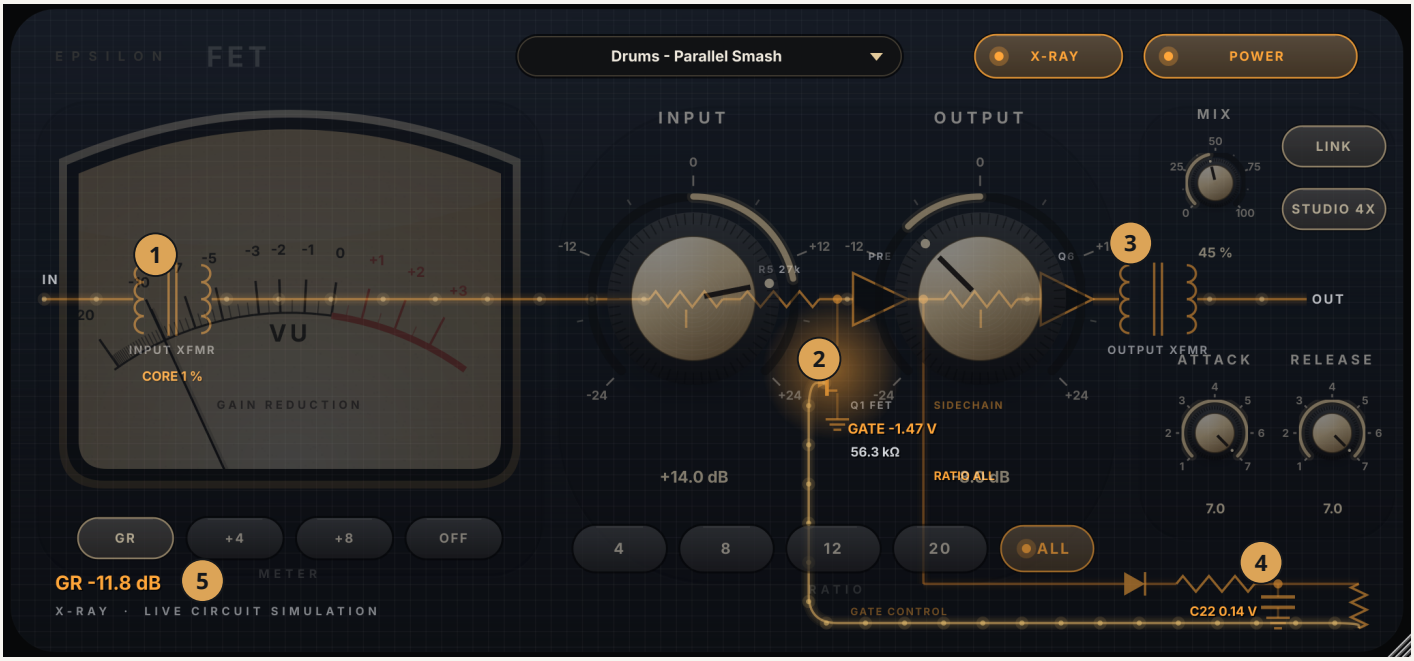
Listen for an even groove. If the mix loses punch or stays pinned after a hit, use less Input or a faster Release.

CATCH A FEW PEAKS

Start from **Peak Limiter** for Ratio 20, Attack 7, Release 5, and Mix 100%. Set Input so it acts mainly on the highest hits. Set Output with the host peak meter. Use a true-peak limiter for a delivery ceiling.

X-RAY and signal flow

X-RAY starts off. Click it to reveal the live circuit model. The same knobs remain usable through the display.



MARKER	WHAT YOU ARE SEEING
1 Input transformer	The incoming signal enters the modeled transformer before the Input pot. The core readout shows modeled flux relative to its saturation knee.
2 Q1 and preamp	Input drives the FET through R5. Q1 changes the attenuation. The display shows the modeled gate voltage and channel resistance.
3 Q6 and output	The preamp feeds the Output pot, then the class-A output stage and output transformer. The wet signal then returns to the host rate for Mix.
4 Detector and C22	The sidechain taps the preamp before Output. Ratio and the diode network feed C22. Attack controls its charging and Release controls its recovery. The resulting voltage controls the FET.
5 GR readout	The numeric value tracks modeled wet-path attenuation. Turn Input up or down while watching the readout and C22 change.

USE IT AS A VISUAL LESSON

Loop a few hits, compare Release 2 and 7, and watch the C22 readout recover. X-RAY changes the view only. Turn it off during CPU-heavy sessions to reduce redraw work.

Troubleshooting

Start with one Epsilon FET instance on one Mixer track so you can isolate routing, level, and CPU behavior.

WHAT YOU SEE OR HEAR	WHAT TO CHECK
FL Studio cannot find it	Confirm the whole Epsilon FET.vst3 bundle is in the VST3 folder and run a full plugin scan with Verify plugins. If your setup scans the system location, try /Library/Audio/Plug-Ins/VST3/ and restart FL Studio. Keep one current copy per format.
macOS blocks the app	Use the ZIP from the successful build and read the warning. For a trusted app with an unidentified developer prompt, follow the Apple instructions for opening apps. If the file is reported damaged, download a fresh copy.
Needle moves, little audible change	Check that Power is lit and Mix is above 0%. GR can still move on the wet circuit while you hear dry audio. Set Mix to 100% temporarily and compare at matched loudness.
GR barely moves	Select GR mode, confirm audio is routed into this Mixer insert, and raise Input. Try Ratio 4 to get compression at a lower level. +4 and +8 show output level.
808 sounds rough or thin	Back down Input, move Release toward 3 or 4, and use a smaller wet blend. Reduce level before Epsilon FET if the input transformer is being driven hard.
Stereo center shifts	Turn Link on. Compare with Power off. If a very loud kick or side hit is ducking the whole bus, ease Input and recheck the loudest section.
Audio crackles or UI lags	Turn X-RAY off, switch to Eco 2x with playback stopped, and close the editor. Try a larger FL Studio audio buffer. If the footer says SLOW UNOPTIMISED BUILD, use the compiled Release ZIP.
Level jumps on bypass	Set Output to match Power on and off over the same loop. Look at the Mixer peak meter as well as the VU. After changing Mix, match loudness again.

RECORD THE EXACT SYMPTOM

For a reproducible problem, note your macOS and FL Studio versions, sample rate, buffer size, format, and Quality setting. Save a short loop and the preset or control values.

Measured behavior

The successful GitHub Actions run compiled universal VST3, AU, and standalone builds. It ran processor tests and headless pluginval level 5 on Intel macOS.

INTEL PROCESSOR CHECK	OBSERVED RESULT
Latency at 48 kHz	6 reported samples and 6 measured samples in the default Studio test (0.125 ms).
Quiet 1 kHz sine	-0.004 dB measured gain at -40 dBFS in the Init preset.
Parallel blend	0.44 dB worst deviation at the measured bands in the 50% Mix test.
Buffer sizes	Identical output for 32, 256, and 1024 sample blocks on the tested signal.
Aliasing test	The 3, 9, and 12 kHz components were below -100 dBFS in the 15 kHz heavy-compression test.
State and automation	Preset state round trip and fast knob automation checks passed.

CPU on the Intel runner

One stereo instance at 44.1 kHz, 64 samples per block. Percentages are of the available processing time for that block.

QUALITY	MEDIAN BLOCK BUDGET	99.9TH PERCENTILE
Studio 4x	28.9%	59.3%
Eco 2x	15.3%	22.7%

What the model represents

The compressor models a Rev D family feedback FET circuit with fitted detector and timing values. The source measurements found slopes near 4.5:1, 12.2:1, 15.3:1, and 23.9:1 for the four numbered buttons over the tested range. Those measurements describe this model. The hardware settings are nominal.

NEXT LISTENING CHECK

The tests cover the compiled plugin and selected signals on the runner. Check loading, playback, recall, and CPU in an actual FL Studio project on your Mac.

GitHub build, logs, and artifact | Source bundle and original engine measurements